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1. Create a file named test1.txt in your home directory.

```
root@localhost:/home# touch test1.txt
root@localhost:/home#
```

2. Create a directory named linux_practice in /home.

```
root@localhost:/home# mkdir linux_practice
root@localhost:/home#
```

3. List all files in the current directory with detailed information.

```
root@localhost:/home# ls -l
total 4
drwx-----. 15 akhil      akhil      4096 Mar 28 01:52 akhil
drwx-----.  3 alice      alice       78 Mar 27 11:56 alice
drwx-----.  3 backupuser backupuser  78 Mar 27 12:21 backupuser
drwx-----.  3 contactor1 contactor1  78 Mar 27 12:49 contactor1
drwx-----.  3 database1  database1  78 Mar 27 12:49 database1
drwx-----.  3 database2  database2  78 Mar 27 12:49 database2
drwx-----.  3 jhon      jhon       78 Mar 27 11:56 jhon
drwxr-xr-x.  2 root      root        6 Mar 28 01:55 linux_practice
drwx-----.  3 operator1  operator1  78 Mar 27 12:49 operator1
drwx-----.  3 rahul      rahul      78 Mar 27 11:52 rahul
-rw-r--r--.  1 root      root        0 Mar 28 01:54 test1.txt
root@localhost:/home#
```

4. Change permissions of test1.txt to read, write, execute for owner and read for others.

```
root@localhost:/home# chmod 744 test1.txt
```

5. Create a user named alex with home directory /home/custom_alex.

```
root@localhost:/home# useradd alex
```

6. Create a group named developers.

```
root@localhost:/home# groupadd developers
root@localhost:/home#
```

7. Add user alex to developers group.

```
root@localhost:/home# usermod -aG developers alex
```

8. Copy test1.txt to linux_practice directory.

```
root@localhost:/home# cp test1.txt linux_practice
root@localhost:/home# ls linux_practice
test1.txt
root@localhost:/home#
```

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9. Move test1.txt to /tmp directory.

```
root@localhost:/home# mv test1.txt /tmp
```

10. Display the content of a file using a command.

```
root@localhost:/home# cat linux_practice/test1.txt
```

11. Create an empty file notes.txt using a command.

```
root@localhost:/home# touch notes.txt  
root@localhost:/home#
```

12. Display last 10 lines of system.log file.

```
root@localhost:/home# tail -10 system.log
```

13. Create multiple directories dir1, dir2, dir3 in one command.

```
root@localhost:/home# mkdir dir1 dir2 dir3
```

14. Remove a directory named dir3.

```
root@localhost:/home# rmdir dir3
```

```
root@localhost:/home# rm -rf dir3
```

15. Rename a file old.txt to new.txt.

```
root@localhost:/home# touch old.txt  
root@localhost:/home# mv old.txt new.txt
```

16. Display current logged-in user.

```
root@localhost:/home# whoami  
root
```

17. Show present working directory.

```
root@localhost:/home# pwd  
/home
```

18. Find location of ls command.

```
root@localhost:/bin# cd  
root@localhost:~# cd /bin  
root@localhost:/bin#
```

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```
lpstat.cups
ls
lsattr
```

19. Display disk usage of current directory.

```
root@localhost:/home# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0   20G  0 disk
├─sda1               8:1    0    1M  0 part
├─sda2               8:2    0    1G  0 part /boot
├─sda3               8:3    0   19G  0 part
│   ├─cs-root        253:0    0   17G  0 lvm  /
│   └─cs-swap        253:1    0    2G  0 lvm  [SWAP]
└─sr0                11:0    1 1024M  0 rom
sr1                  11:1    1   7.1G  0 rom  /run/media/akhil/CentOS-Stream-10-BaseOS-x86_64
```

20. Compress a file using tar command.

21. Create directory /home/project with subdirectories docs and scripts.

```
root@localhost:/home# mkdir /home/projects
root@localhost:/home# cd projects
root@localhost:/home/projects# mkdir docs scripts
root@localhost:/home/projects# ls
docs  scripts
root@localhost:/home/projects#
```

22. Create readme.txt inside docs and add text.

```
root@localhost:/home/projects# vim docs/readme.txt
root@localhost:/home/projects#
```

24. Set permissions: owner full, group read+execute, others none for scripts directory.

```
root@localhost:/home/projects# chmod 750 scripts
root@localhost:/home/projects# ls -l
total 0
drwxr-xr-x. 2 root root 24 Mar 28 02:25 docs
drwxr-x---. 2 root root  6 Mar 28 02:20 scripts
```

25. Copy all files from docs to scripts.

```
root@localhost:/home/projects# cp docs/* scripts
root@localhost:/home/projects# ls scripts
readme.txt
```

26. Move test.sh from /tmp to scripts directory.

```
root@localhost:~# mv /tmp/test.sh /projects/scripts
```

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27. Create user tester with /bin/bash shell.

```
root@localhost:~# useradd -s /bin/bash tester
```

28. Create group qa_team and add tester to it.

```
root@localhost:~# groupadd qa_team  
root@localhost:~# usermod -aG qa_team tester
```

30. Find all .txt files in /home/project.

```
root@localhost:/home# grep *.txt projects
```